

COMP0078: Supervised Learning

Collected Problem Sheet with Solutions

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1. Introduction to Supervised Learning

Fundamentals of supervised learning: nearest-neighbour methods, Bayes-optimal estimators, and linear regression basics.

Problem 1.1.

“For the K -nearest neighbour algorithm, larger K values will tend to lead to overfitting.” True or False? Explain your answer.

Solution 1.1.

False. Larger k creates more regular, smoother decision boundaries, which could possibly lead to *underfitting*. $\square$

Problem 1.2.

First, give Cover’s bound on the Bayes error of the 1-nearest neighbour algorithm as the number of examples goes to infinity. Second, give an example where the 1-nearest neighbour algorithm achieves the Bayes error as the number of examples goes to infinity, and explain your reasoning.

Solution 1.2.

Two possibilities are when $p^* = 1/C$ (each class is equally likely) or $p^* = 1$ (deterministic case, Bayes error = 0). In the former case $\sum_c P(c | x)(1 - P(c | x)) = 1 - \frac{1}{C}$ and $1 - p^* = 1 - \frac{1}{C}$. In the latter case $\sum_c P(c | x)(1 - P(c | x)) = 1 - p^* = 0$. $\square$

Problem 1.3.

How can we choose k to ensure that k -NN makes an unambiguous decision in the two-class case, provided the test point is not equidistant to any pair of training points? What values of k give an unambiguous decision under similar assumptions for the 3-class case?

Solution 1.3.

2-class case: k must be odd (if $k = 2r$ is even we can have r points from each class, giving a tie).

3-class case: $k = 1$ (if $k = 2r$ is even the same reasoning as the 2-class case applies; if $k = 2r + 1$ for $r \geq 1$ then we can have one point labelled from the third class and the $2r$ remaining points split evenly between the other two classes). $\square$

Problem 1.4.

One of the drawbacks of the nearest-neighbour algorithm is that we must retain all of the training data. Describe a situation where a training point can be removed without affecting the resulting 1-NN classification for any test point in the input space.

Solution 1.4.

Construct a Voronoi diagram: for each training point x we construct a region R_x such that every point $y \in R_x$ is closer to x than to all other training points. If a region R_x of a point x with label ℓ has boundaries *only* with regions of points sharing the same label ℓ , then we can remove x without affecting the 1-NN classification. After removal, every point in R_x will be closest to one of the neighbours of R_x , which all share the same label. $\square$

Problem 1.5.

In linear regression it is common to transform the training data $(x_1, y_1), \dots, (x_m, y_m)$ to $((x_1, 1), y_1), \dots, ((x_m, 1), y_m)$. What is the motivation for this procedure?

Solution 1.5.

To incorporate a bias term. $\square$

Problem 1.6.

Given a dataset $\{(x_1, y_1), \dots, (x_m, y_m)\}$ where $x \in \mathbb{R}^n$ and $y \in \{1, \dots, K\}$. How can linear regression (least squares) be used or adapted to tackle such a multi-class classification problem?

Solution 1.6.

Using one-vs-one (OvO) or one-vs-rest (OvR) strategies. For OvR, there are K classifiers; each binary classifier is fed $+1$ for class k and -1 for all others. Predictions are combined by choosing the class with the highest score $\langle w, x \rangle - \theta$. For OvO, each classifier is fed one of the $\binom{K}{2}$ combinations of two classes, and the final prediction is by majority vote. $\square$

Problem 1.7.

Given $Y \subset \mathbb{R}$ and $|Y| < \infty$, $|X| < \infty$ and a probability mass function $p: X \times Y \rightarrow \mathbb{R}$, find $f^* := \arg \min_f \mathbb{E}[|x - f(y)|]$ (Bayes estimator with L_1 loss).

Solution 1.7.

The Bayes estimator is

$$f^* = \arg \min_f \mathbb{E}[L(y, f(x))] = \sum_{x \in X} \left(\sum_{y \in Y} |y - f(x)| p(y | x) \right) p(x).$$

Fix $x = x_0$ and let $\hat{y} = f^*(x_0)$. Taking the derivative with respect to $\hat{y}$ and setting to zero:

$$\begin{aligned} 0 &= \frac{\partial}{\partial \hat{y}} \sum_{y \in Y} |y - \hat{y}| p(y | x_0) \\ &= \sum_{\substack{y \in Y \\ y < \hat{y}}} p(y | x_0) - \sum_{\substack{y \in Y \\ y \geq \hat{y}}} p(y | x_0) \\ &= P(Y < \hat{y} | X = x_0) - P(Y \geq \hat{y} | X = x_0). \end{aligned}$$

Hence $P(Y \geq \hat{y} | X = x_0) = P(Y < \hat{y} | X = x_0) = \frac{1}{2}$. By definition of the median, the Bayes estimator $\hat{y}(x)$ is the **median** of $Y | X = x$. $\square$

2. Features, Kernels and Regularisation

Convexity fundamentals, primal/dual formulations for ridge regression, kernel construction and closure properties, and Mercer's theorem.

2.1 Convexity

Problem 2.1.

If f and g are convex then $f + g$ is convex.

Solution 2.1.

For $p, q \in X$ and $\alpha \in (0, 1)$:

$$\begin{aligned}(f + g)(\alpha p + (1 - \alpha)q) &= f(\alpha p + (1 - \alpha)q) + g(\alpha p + (1 - \alpha)q) \\ &\leq \alpha f(p) + (1 - \alpha)f(q) + \alpha g(p) + (1 - \alpha)g(q) \\ &= \alpha(f(p) + g(p)) + (1 - \alpha)(f(q) + g(q)).\end{aligned}$$

□

Problem 2.2.

If f is convex and g is affine then $f(g(\cdot))$ is convex.

Solution 2.2.

Since g is affine, $g(x) = ax + b$ for some $a, b \in \mathbb{R}$. Then:

$$\begin{aligned}(f \circ g)(\alpha p + (1 - \alpha)q) &= f(a(\alpha p + (1 - \alpha)q) + b) \\ &= f(\alpha(ap + b) + (1 - \alpha)(aq + b)) \\ &\leq \alpha f(ap + b) + (1 - \alpha)f(aq + b) \\ &= \alpha f(g(p)) + (1 - \alpha)f(g(q)).\end{aligned}$$

□

Problem 2.3.

Suppose M is a symmetric matrix. Then M is PSD if and only if $f(x) = x^\top Mx$ is convex.

Solution 2.3.

($\Rightarrow$) If M is PSD, the Hessian of f is $\frac{\partial^2}{\partial x^2} x^\top Mx = 2M$, which is PSD. By the second-order condition (Problem 2.5), f is convex.

($\Leftarrow$) If f is convex, it attains its minimum when $\nabla f(x) = 2Mx = 0$, i.e. at $x = 0$. So $x^\top Mx = f(x) \geq f(0) = 0$ for all x , proving M is PSD. □

Problem 2.4.

A level set of a convex function is convex.

Solution 2.4.

Let $L = \{x : f(x) \leq c\}$ be a level set of convex f . For $p, q \in L$ we have $f(p) \leq c$ and $f(q) \leq c$. Then:

$$f(\alpha p + (1 - \alpha)q) \leq \alpha f(p) + (1 - \alpha)f(q) \leq \alpha c + (1 - \alpha)c = c,$$

so $\alpha p + (1 - \alpha)q \in L$. □

Problem 2.5.

If $X \subseteq \mathbb{R}^n$ is convex and the Hessian $H(x)$ is PSD for all $x \in X$, then f is convex.

Solution 2.5.

By Taylor's Theorem, there exists $t \in (0, 1)$ such that

$$f(x) = f(a) + (x - a)^\top \nabla f(a) + \frac{1}{2}(x - a)^\top H(a + t(x - a))(x - a).$$

Since H is PSD, $(x - a)^\top H(\cdot)(x - a) \geq 0$, giving $f(x) \geq f(a) + (x - a)^\top \nabla f(a)$.

For $p, q \in X$, $\alpha \in [0, 1]$, let $a = \alpha p + (1 - \alpha)q$. Then:

$$\begin{aligned} \alpha f(p) + (1 - \alpha)f(q) &\geq \alpha [f(a) + (p - a)^\top \nabla f(a)] + (1 - \alpha) [f(a) + (q - a)^\top \nabla f(a)] \\ &= f(a) + \nabla f(a)^\top [\alpha(1 - \alpha)(p - q) + (1 - \alpha)\alpha(q - p)] \\ &= f(a) = f(\alpha p + (1 - \alpha)q). \end{aligned}$$

□

2.2 Primal vs. Dual Formulation**Problem 2.6.**

When is it advantageous to compute the linear regression solution via the primal vs. the dual?

Solution 2.6.

The matrix X is of size $m \times n$. The primal solution $w = (X^\top X + \lambda I_n)^{-1} X^\top y$ inverts an $n \times n$ matrix. The dual solution $w = \sum_{i=1}^m \alpha_i x_i$ where $\alpha = (X X^\top + \lambda I_m)^{-1} y$ inverts an $m \times m$ matrix. If $n > m$ (more dimensions than samples), use the dual. If $m > n$, use the primal. □

2.3 Kernel Construction and Validity**Problem 2.7.**

Given $K(x, t) = (1 + \langle x, t \rangle)^2$, find a feature map $\varphi: \mathbb{R}^2 \rightarrow \mathbb{R}^6$.

Solution 2.7.

Expanding:

$$\begin{aligned} K(x, t) &= (1 + x_1 t_1 + x_2 t_2)^2 \\ &= 1 + x_1^2 t_1^2 + x_2^2 t_2^2 + 2x_1 t_1 + 2x_2 t_2 + 2x_1 x_2 t_1 t_2. \end{aligned}$$

So $\varphi(x) = (1, x_1^2, x_2^2, \sqrt{2}x_1, \sqrt{2}x_2, \sqrt{2}x_1x_2)^\top$ gives $K(x, t) = \varphi(x)^\top \varphi(t)$. $\square$

Problem 2.8.

Determine whether the following are valid kernels:

- (a) $K(x, t) = x^\top D t$, $D = \text{diag}(1, 5)$
- (b) $K(x, t) = x^\top D t$, $D = \begin{pmatrix} -1 & 2 \\ 2 & 4 \end{pmatrix}$
- (c) $K(x, t) = \exp(x_1 t_1)$
- (d) $K(x, t) = x^\top t - (x^\top t)^2$
- (e) Prove closure under (i) $K(Ax, At)$ and (ii) $f(x)K(x, t)f(t)$.

Solution 2.8.

(a) Valid kernel. $K(x, t) = x_1 t_1 + 5x_2 t_2 = \varphi(x)^\top \varphi(t)$ where $\varphi(x) = (x_1, \sqrt{5}x_2)^\top$.

(b) Not a valid kernel. For $x = (1, 0)^\top$: $K(x, x) = -1 < 0$, so K is not PSD.

(c) Valid kernel. Using $e^u = \sum_{n=0}^{\infty} \frac{u^n}{n!}$:

$$K(x, t) = \sum_{n=0}^{\infty} \frac{x_1^n t_1^n}{n!} = \varphi(x)^\top \varphi(t) \quad \text{where } \varphi(x) = (1, x_1, x_1^2/\sqrt{2!}, \dots)^\top.$$

(d) Not a valid kernel. For $x = (\sqrt{2}, 0)^\top$: $K(x, x) = 2 - 4 = -2 < 0$.

(e-i) Let $\tilde{x}_i := Ax_i$. Since K is a kernel, $\sum_{i,j} c_i c_j K(\tilde{x}_i, \tilde{x}_j) \geq 0$, i.e. $\sum_{i,j} c_i c_j K(Ax_i, Ax_j) \geq 0$.

(e-ii) Let $\tilde{c}_i := f(x_i)c_i$. Since K is a kernel, $\sum_{i,j} \tilde{c}_i \tilde{c}_j K(x_i, x_j) = \sum_{i,j} c_i c_j f(x_i)K(x_i, x_j)f(x_j) \geq 0$. $\square$

Problem 2.9.

[Hard]

Does the Gaussian kernel $K(x, z) = e^{-\|x-z\|^2}$ admit a finite-dimensional feature map?

Solution 2.9.

No. Suppose for contradiction $\exists \varphi: \mathbb{R}^n \rightarrow \mathbb{R}^d$ with $K(x, z) = \sum_{i=1}^d \varphi_i(x)\varphi_i(z)$.

Consider points $x_\ell := (\ell, 0, \dots, 0) \in \mathbb{R}^n$ for $\ell \in \mathbb{N} \setminus \{0\}$. For $\ell \neq h$, the angle θ between $\varphi(x_\ell)$ and $\varphi(x_h)$ satisfies

$$\cos \theta = \frac{K(x_\ell, x_h)}{\sqrt{K(x_\ell, x_\ell)K(x_h, x_h)}} = e^{-|\ell-h|} \leq e^{-1}.$$

So $\theta \geq \arccos(e^{-1}) > 0$. This gives infinitely many vectors in $\mathbb{R}^d$ with pairwise angle bounded away from zero, which is impossible in a finite-dimensional space. Contradiction. $\square$

2.4 Kernels on Structured Domains

Problem 2.10.

Argue that every $x \in X$ has an additive inverse $-x \in X$ with $x + (-x) = 0$.

Solution 2.10.

$0 = 0x = (1 - 1)x = 1x + (-1)x = x + (-1)x$. So the additive inverse of x is $(-1)x \in X$. $\square$

Problem 2.11.

Prove $K(A, B) = 2^{|A \cap B|}$ is a kernel on $2^X \times 2^X$.

Solution 2.11.

Define features indexed by $C \in 2^X$: $\varphi_C(A) = [C \subseteq A]$ (Iverson bracket). Then:

$$\sum_{C \in 2^X} \varphi_C(A) \varphi_C(B) = \sum_{C \in 2^X} [C \subseteq A][C \subseteq B] = \sum_{C \in 2^X} [C \subseteq A \cap B] = |2^{A \cap B}| = 2^{|A \cap B|} = K(A, B).$$

$\square$

Problem 2.12.

Min kernel. (a) Prove $\min(x, t)$ is a kernel on $[0, \infty)$. (b) Find a feature map.

Solution 2.12.

(a) Consider the inner product space F of absolutely continuous functions $f: [0, \infty) \rightarrow \mathbb{R}$ with $f(0) = 0$ and $\int_0^\infty (f'(u))^2 du < \infty$, equipped with $\langle f, g \rangle = \int_0^\infty f'(u)g'(u) du$. Define $\Phi_x(u) = \min(u, x)$. Then $\Phi'_x(u) = \mathbf{1}(u \leq x)$ and:

$$\langle \Phi_x, \Phi_t \rangle = \int_0^\infty \mathbf{1}(u \leq x) \mathbf{1}(u \leq t) du = \int_0^{\min(x, t)} 1 du = \min(x, t).$$

(b) By Mercer's Theorem on $[0, M]$, solving the eigenvalue problem $\lambda e''(x) = -e(x)$ with $e(0) = 0$ and $e'(M) = 0$ yields

$$\lambda_\ell = \frac{M^2}{(\ell + \frac{1}{2})^2 \pi^2}, \quad e_\ell(x) = \sqrt{\frac{2}{M}} \sin\left(\frac{(\ell + \frac{1}{2})\pi}{M} x\right).$$

The feature map is $\varphi_\ell(x) = \sqrt{\lambda_\ell} e_\ell(x) = \frac{\sqrt{2}M}{(\ell + \frac{1}{2})\pi} \sin\left(\frac{(\ell + \frac{1}{2})\pi}{M} x\right)$ and $\min(x, t) = \sum_{\ell=0}^\infty \varphi_\ell(x) \varphi_\ell(t)$. $\square$

3. Support Vector Machines

Hard- and soft-margin SVM formulations, Lagrangian duality, KKT conditions, and the connection between the hinge loss and regularisation.

3.1 Geometric Intuition

Problem 3.1.

Describe the hard margin SVM criterion, illustrating with a diagram.

Solution 3.1.

The criterion is to choose the separating hyperplane (an affine plane with all positive examples on one side and all negative examples on the other) that **maximises the distance of the nearest example to it**. The nearest examples are the *support vectors*, and the distance from the hyperplane to the nearest example on either side is the *margin*. $\square$

Problem 3.2.

For hard-margin SVM, how does removing (a) a support vector, (b) a non-support vector affect the margin?

Solution 3.2.

- (a) If we remove a support vector, the maximum margin either *stays the same* or *increases*; it never decreases.
 (b) If we remove a non-support vector, the optimal hyperplane and its margin remain *unchanged*. $\square$

3.2 Soft-Margin SVM

Problem 3.3.

For $\min_{w, \xi} \|w\| + C \sum_{i=1}^m \xi_i$ subject to $y_i \langle w, x_i \rangle \geq 1 - \xi_i$, $\xi_i \geq 0$: identify the regulariser and loss.

Solution 3.3.

Rewriting equivalently:

$$\min_w \sum_{i=1}^m \max(0, 1 - y_i \langle w, x_i \rangle) + C^{-1} \|w\|.$$

The loss function is the **hinge loss** $V_{\text{hinge}}(y_i, \langle w, x_i \rangle) = \max(0, 1 - y_i \langle w, x_i \rangle)$, and $\|w\|$ is the **regulariser** with regularisation coefficient C^{-1} . $\square$

3.3 Lagrangian Duality and KKT Conditions

Problem 3.4.

Argue that $P_1: \min_w \frac{1}{2}w^\top w$ s.t. $y_i w^\top x_i \geq 1$ has a unique solution and describe its geometric meaning.

Solution 3.4.

Existence: Since the data is strictly linearly separable through the origin, there exists some feasible $\bar{w}$. Adding the constraint $\|w\| \leq \|\bar{w}\|$ makes the feasible set compact, guaranteeing a solution.

Uniqueness: The objective $\frac{1}{2}w^\top w$ is strictly convex (Hessian = I). The feasible set is the intersection of half-spaces (convex) intersected with a ball (convex), hence convex and non-empty. A strictly convex function on a non-empty convex compact set has a unique minimiser.

Geometric meaning: The solution is the separating hyperplane through the origin that maximises the margin. $\square$

Problem 3.5.

Show $w = \sum_{i=1}^m c_i y_i x_i$ with $c_i \geq 0$ using Lagrange multipliers.

Solution 3.5.

The Lagrangian is $L(w, c) = \frac{1}{2}w^\top w - \sum_{i=1}^m c_i (y_i w^\top x_i - 1)$. Setting $\nabla_w L = 0$:

$$0 = w - \sum_{i=1}^m c_i y_i x_i \implies w = \sum_{i=1}^m c_i y_i x_i.$$

The dual formulation requires $c \geq 0$. $\square$

Problem 3.6.

Show that c solves P_2 and that $w^\top w = \sum_{i=1}^m c_i$.

Solution 3.6.

Substituting $w = \sum_i c_i y_i x_i$ into the Lagrangian:

$$\begin{aligned} L(w, c) &= \frac{1}{2} \sum_{i,j} c_i c_j y_i y_j x_i^\top x_j - \sum_{i,j} c_i c_j y_i y_j x_i^\top x_j + \sum_i c_i \\ &= -\frac{1}{2} \sum_{i,j} c_i c_j y_i y_j x_i^\top x_j + \sum_i c_i. \end{aligned}$$

So c maximises $P_2: \max_{c \geq 0} -\frac{1}{2} \sum_{i,j} c_i c_j y_i y_j x_i^\top x_j + \sum_i c_i$.

For the KKT conditions: either $c_i = 0$ or $\sum_j c_j y_i y_j x_i^\top x_j = 1$. Then:

$$w^\top w = \sum_{i,j} c_i c_j y_i y_j x_i^\top x_j = \sum_{i: c_i > 0} c_i \cdot 1 = \sum_{i=1}^m c_i.$$

$\square$

4. Tree-Based Learning and Ensemble Methods

Decision/regression trees via greedy partitioning, bagging and random forests for variance reduction, and boosting (AdaBoost) for bias reduction.

4.1 Regression Trees

Problem 4.1.

Find the regression tree for the dataset $\{((1, 1), 9), ((1, 2), -4), ((1, 3), 2), ((2, 2), 4), ((2, 3), 2)\}$ using greedy recursive partitioning with square error.

Solution 4.1.

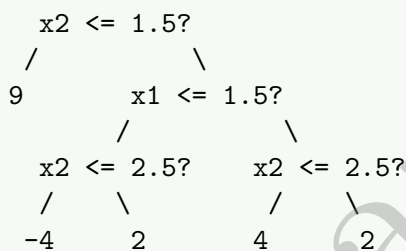
First split: Compare all axis-aligned splits. $A_1: x_2 \leq 1.5$ gives error 36; $A_2: x_2 \leq 2.5$ gives error 86; $A_3: x_1 \leq 1.5$ gives error ≈ 86.7 . Choose A_1 ($x_2 \leq 1.5$).

Second split (in region $x_2 > 1.5$): $B_1: x_2 \leq 2.5$ gives error 32; $B_2: x_1 \leq 1.5$ gives error 20. Choose B_2 .

Third split (region $x_2 > 1.5 \wedge x_1 \leq 1.5$): only split is $x_2 \leq 2.5$, giving leaves -4 and 2 .

Fourth split (region $x_2 > 1.5 \wedge x_1 > 1.5$): only split is $x_2 \leq 2.5$, giving leaves 4 and 2 .

Final tree:



□

4.2 Ensemble Methods: Bagging and Boosting

Problem 4.2.

Explain how a random forest is constructed and its motivation.

Solution 4.2.

The random forest method encourages each decision tree to be uncorrelated:

1. Draw M bootstrap samples from the training data (sample with replacement).
2. For each sample, consider a random subset of features ($n_{\text{tree}} \approx \sqrt{n}$ or $\log n$).
3. Build a decision tree on that sample using only those features.
4. Repeat N times to generate N trees.
5. Predict by averaging (regression) or majority vote (classification).

Both bootstrap sampling and random feature selection decorrelate the trees. Decision trees are high-variance, so bagging (which reduces variance) is effective. □

Problem 4.3.

Explain how bagging and boosting ensembles differ structurally.

Solution 4.3.

Bagging: Each sample has equal weight; each learner sees a random *subset* of the data. Bias of each learner equals the ensemble's bias. Motivation: *variance reduction*.

Boosting: Places more weight on samples previous models got wrong. Each learner sees *all* data with variable sample weights. Learners are added sequentially to reduce *bias*. $\square$

4.3 AdaBoost**Problem 4.4.**

AdaBoost initialisation and update.

Solution 4.4.

(a) Equal weighting: $D_0(i) = 1/m$ for all i .

(b) The update is:

$$D_{t+1}(i) = \frac{D_t(i) e^{-\alpha_t y_i h_t(x_i)}}{Z_t}$$

where $\alpha_t = \frac{1}{2} \ln\left(\frac{1-\varepsilon_t}{\varepsilon_t}\right)$, $\varepsilon_t = \sum_{i=1}^m D_t(i) \mathbb{I}[h_t(x_i) \neq y_i]$, and $Z_t = \sum_{i=1}^m D_t(i) e^{-\alpha_t y_i h_t(x_i)}$.

(c) For correctly classified examples ($y_i h_t(x_i) = 1$): $D_{t+1}(i) \propto D_t(i) e^{-\alpha_t}$, so the weight *decreases*. For incorrectly classified examples ($y_i h_t(x_i) = -1$): $D_{t+1}(i) \propto D_t(i) e^{\alpha_t}$, so the weight *increases*. This focuses subsequent learners on harder examples. $\square$

Problem 4.5.

How does AdaBoost choose a weak learner at stage t ?

Solution 4.5.

Any hypothesis satisfying $E_t = \sum_{i=1}^m D_t(i) \mathbb{I}[h_t(x_i) \neq y_i] < \frac{1}{2}$ will work. The optimal weak learner is the hypothesis that *minimises* E_t . $\square$

Problem 4.6.

Bound the training error in terms of $Z_1, \dots, Z_T$.

Solution 4.6.

First, $\mathbb{I}[\text{sign}(f_T(x_i)) \neq y_i] \leq \exp(-y_i f_T(x_i))$ since: if correct, LHS = 0; if wrong, $y_i f_T(x_i) \leq 0$ so RHS ≥ 1 .

From the AdaBoost update, $D_{T+1}(i) = \frac{1}{m} \frac{\prod_{t=1}^T e^{-\alpha_t y_i h_t(x_i)}}{\prod_{t=1}^T Z_t}$. Therefore:

$$\frac{1}{m} \sum_{i=1}^m \exp(-y_i f_T(x_i)) = \frac{1}{m} \sum_{i=1}^m \prod_{t=1}^T e^{-\alpha_t y_i h_t(x_i)} = \sum_{i=1}^m D_{T+1}(i) \prod_{t=1}^T Z_t = \prod_{t=1}^T Z_t.$$

$\square$

Problem 4.7.

(a) Is H_{ds} weakly-universal? (b) Is H_{SS} weakly-universal?

Solution 4.7.

(a) H_{ds}^1 (for $n = 1$) is weakly-universal. Given an ordered generic dataset, take $z = x_m$; if the error of h_{1,x_m} is exactly 0.5, shift to $z = (x_{m-1} + x_m)/2$, which classifies x_m differently, breaking the tie. If the error exceeds 0.5, negate the stump.

For $n > 1$, H_{ds} is *not* weakly-universal. Counterexample: the XOR dataset $\{(0, 0, -1), (0, 1, +1), (1, 0, +1), (1, 1, -1)\}$ with uniform weights. Every decision stump classifies exactly 2 correctly and 2 wrongly, giving error exactly 0.5.

(b) H_{SS} is weakly-universal. The proof follows the same structure as the $n = 1$ decision stump case. Take z as the concatenation of all strings; if error is exactly 0.5, remove the longest string x_k from the concatenation so that $x_k \not\subseteq z$ while $x_i \subseteq z$ for $i \neq k$, breaking the tie. $\square$

5. Online Learning

Mistake-bounded learning via the Perceptron, competitive analysis via the Weighted Majority algorithm, and lower bounds from adversarial constructions.

5.1 The Perceptron

Problem 5.1.

Give a polynomial time algorithm with mistake bound $M(A) \leq O(n^2)$ for learning k -literal conjunctions.

Solution 5.1.

Use the **Perceptron algorithm**. The mistake bound is $M \leq R^2 \|u\|^2$ where $R^2 = \max_t \|x_t\|^2$. Using de Morgan's Law, represent the conjunction as a disjunction of negated variables. Let S be the set of indices in the conjunction and define:

$$u_i = \begin{cases} -2 & i \in S, \\ 1 & i = n+1, \\ 0 & \text{otherwise,} \end{cases} \quad \varphi(x) = (\bar{x}, 1).$$

Then $R^2 \leq n+1$ and $\|u\|^2 = 4k+1$, giving $M(A) \leq (n+1)(4k+1) = O(n^2)$. $\square$

Problem 5.2.

State the Perceptron convergence theorem and its relation to hard margin SVM.

Solution 5.2.

The number of mistakes made by the Perceptron is $M \leq R^2/\gamma^2$, if there exists a unit vector v ($\|v\| = 1$) satisfying $\langle v, x_t \rangle y_t \geq \gamma$. The term γ is analogous to the SVM margin. $\square$

5.2 Weighted Majority and Regret Bounds

Problem 5.3.

(a) Show c -regret < 2 is impossible without randomisation. (b) Select β for sublinear 2-regret.

Solution 5.3.

(a) Consider two experts giving constant predictions 0 and 1. An adversary forces a mistake every round, so $M_A = T$. But $\min_i M_i \leq T/2$. Hence $M_A \geq 2 \min_i M_i$, so sublinear c -regret for $c < 2$ is impossible.

(b) Starting from $M \leq \frac{\ln(1/\beta)}{\ln(2/(1+\beta))} M_i + \frac{\ln n}{\ln(2/(1+\beta))}$, let $\varepsilon = 1 - \beta$ and use $-\ln(1 - \varepsilon) \leq \varepsilon + \varepsilon^2$ and $-\ln(1 + x) \geq -x$:

$$M \leq 2M_i + 2\varepsilon T + \frac{2}{\varepsilon} \ln n.$$

By AM-GM, the optimal $\varepsilon = \sqrt{\frac{\ln n}{T}}$ gives $M \leq 2M_i + 4\sqrt{T \ln n} + 4 \ln n$. $\square$

Problem 5.4.

Prove any algorithm makes $\Omega(\min(T, \log_2 n))$ mistakes with a perfect expert.

Solution 5.4.

Construct $n = 2^N$ experts via a complete binary prediction tree of depth N . Each path from root to leaf defines an expert's predictions. At each node, the adversary reveals the label that causes the algorithm to err, eliminating at most half the remaining experts. After $N = \log_2 n$ rounds, N mistakes have been forced (or T if $T < N$). Hence the lower bound is $\Omega(\min(T, \log_2 n))$. $\square$

Problem 5.5.

Design a randomised algorithm achieving $M/T \rightarrow \min_i M_i/T$ as $T \rightarrow \infty$.

Solution 5.5.

Use **Randomised Weighted Majority**: at each step, predict $\hat{y}_t = 0$ with probability $\frac{\sum_{i: x_{t,i}=0} w_{t,i}}{W_t}$ and 1 otherwise. Update by multiplying incorrect experts' weights by β . By the same analysis as Problem 5.3(b) but with the $(1 + \varepsilon)$ factor instead of $2(1 + \varepsilon)$:

$$\mathbb{E}[M] \leq M_i + \varepsilon T + \frac{1}{\varepsilon} \ln n.$$

Setting $\varepsilon = \sqrt{\frac{\ln n}{T}}$:

$$\frac{\mathbb{E}[M]}{T} - \frac{M_i}{T} \leq 2\sqrt{\frac{\ln n}{T}} \rightarrow 0 \quad \text{as } T \rightarrow \infty.$$

$\square$

6. Graphs in Machine Learning

Graph-based semi-supervised learning: nearest-neighbour graphs, the graph Laplacian, harmonic energy minimisation, min-cut relaxations, and effective resistance.

6.1 Graph Construction and the Laplacian

Problem 6.1.

Describe the differences between semi-supervised, supervised, and unsupervised learning.

Solution 6.1.

Unsupervised: Learns structure from unlabelled data only.

Supervised: Learns from labelled data to predict labels of unseen data.

Semi-supervised: Uses both labelled and unlabelled data simultaneously to learn structure and predict labels. $\square$

Problem 6.2.

Draw the 2-nearest neighbour graph for $(-2, 0)$, $(-1, -1)$, $(0, 0)$, $(1, 1)$, $(2, 0)$.

Solution 6.2.

Label points $A = (-2, 0)$, $B = (-1, -1)$, $C = (0, 0)$, $D = (1, 1)$, $E = (2, 0)$. The 2-nearest neighbours are: $A \rightarrow \{B, C\}$, $B \rightarrow \{A, C\}$, $C \rightarrow \{B, D\}$, $D \rightarrow \{C, E\}$, $E \rightarrow \{C, D\}$. Edges: $A-B$, $A-C$, $B-C$, $C-D$, $C-E$, $D-E$. $\square$

Problem 6.3.

Give the graph Laplacian.

Solution 6.3.

From the adjacency matrix, $L = D - A$:

$$L = \begin{pmatrix} 2 & -1 & -1 & 0 & 0 \\ -1 & 2 & -1 & 0 & 0 \\ -1 & -1 & 4 & -1 & -1 \\ 0 & 0 & -1 & 2 & -1 \\ 0 & 0 & -1 & -1 & 2 \end{pmatrix}.$$

$\square$

6.2 Min-Cut and Harmonic Labelling

Problem 6.4.

(a) Minimum cut solution at v_3 . (b) Harmonic energy solution at v_3 .

Solution 6.4.

- (a) The label for v_3 can be either $+1$ or -1 ; both yield minimum cut value 2.
 (b) Minimising $S_2 = \sum_{(u,v) \in E} |\psi(u) - \psi(v)|^2$ with $\psi(v_1) = 1$ and $\psi(v_5) = -1$, the energy is minimised when the differences are equal along the path: $\psi(v_3) = 1/3$. $\square$

Problem 6.5.

Find the harmonic labelling of D_m (two m -cliques connected by a bridge) as $m \rightarrow \infty$.

Solution 6.5.

By symmetry and solving $\partial S_2 / \partial \psi(v_i) = 0$, one obtains:

$$\psi(v_j) = \frac{m+2}{m+4} \text{ for } j = 2, \dots, m-1, \quad \psi(v_m) = \frac{m}{m+4},$$

and symmetrically $\psi(v_{m+1}) = -\frac{m}{m+4}$, etc. As $m \rightarrow \infty$: $\psi(v_j) \rightarrow 1$ for $j = 2, \dots, m$ and $\psi(v_j) \rightarrow -1$ for $j = m+1, \dots, 2m$. $\square$

6.3 Advanced: LP Relaxations, Effective Resistance, and Laplacian Kernels**Problem 6.6.****[Hard]**

Prove the min-cut LP relaxation is tight.

Solution 6.6.

Let α be the IP optimum and β the LP optimum. Clearly $\beta \leq \alpha$. Given the LP solution u^* , pick $r \sim \text{Uniform}(0, 1)$ and set $S = \{v : u_v^* < r\}$. This gives a feasible IP solution. The probability that edge (i, j) is cut equals $|u_i^* - u_j^*| = x_{ij}^*$. So the expected cost equals $\sum_{ij} w_{ij} x_{ij}^* = \beta$. Since every realisation has cost $\geq \alpha$, we get $\alpha \leq \beta$. Combined: $\alpha = \beta$. $\square$

Problem 6.7.**[Hard]**

Prove $\max_k L_{kk}^+ \leq \max_{i,j} r_G(i, j)$.

Solution 6.7.

Using $r_G(i, j) = e_i^\top L^+ e_i - 2e_i^\top L^+ e_j + e_j^\top L^+ e_j$ and $L^+ \mathbf{1} = 0$:

$$\frac{1}{n} \sum_{j=1}^n r_G(i, j) = L_{ii}^+ + \frac{1}{n} \text{tr}(L^+) \geq L_{ii}^+$$

since $\text{tr}(L^+) \geq 0$ (L^+ is PSD). Also $\frac{1}{n} \sum_j r_G(i, j) \leq \max_{i,j} r_G(i, j)$. Taking max over k : $\max_k L_{kk}^+ \leq \max_{i,j} r_G(i, j)$. $\square$

Problem 6.8.**[Hard]**

Prove $L_{vv}^+ = R(v)/n - R_{\text{tot}}/n^2$.

Solution 6.8.

From the definitions:

$$\frac{R(v)}{n} = \frac{1}{n} \sum_u (L_{vv}^+ + L_{uu}^+ - 2L_{uv}^+) = L_{vv}^+ + \frac{1}{n} \sum_u L_{uu}^+.$$

$$\frac{R_{\text{tot}}}{n^2} = \frac{1}{2n^2} \sum_{u,v} (L_{uu}^+ + L_{vv}^+) = \frac{1}{n} \sum_u L_{uu}^+.$$

Subtracting: $R(v)/n - R_{\text{tot}}/n^2 = L_{vv}^+.$

□

Problem 6.9.**[Hard]**

Prove $\sum_{(i,j) \in G} r_G(i,j) = n - 1$ for unweighted connected graphs.

Solution 6.9.

Define the incidence matrix $C \in \mathbb{R}^{|E| \times |V|}$ so that $L = C^\top C$. Let $\Gamma = CL^+C^\top$. Then $\Gamma_{\ell,\ell} = r_G(i,j)$ for edge ℓ connecting i,j . Using $\text{tr}(AB) = \text{tr}(BA)$:

$$\sum_{(i,j) \in G} r_G(i,j) = \text{tr}(\Gamma) = \text{tr}(CL^+C^\top) = \text{tr}(L^+C^\top C) = \text{tr}(L^+L).$$

The eigenvalues of L^+L are 1 (with multiplicity $n-1$) and 0 (multiplicity 1, for the connected graph). So $\text{tr}(L^+L) = n - 1$.

□

Problem 6.10.**[Hard]**

Prove the equality vs. inequality constraint result for Laplacian kernels.

Solution 6.10.

Let $K = (L + \varepsilon I)^{-1}$. Suppose for contradiction there exists v in the inequality-constrained set with $v^\top K^{-1}v$ strictly less than the equality-constrained minimum. Let $\iota = \arg \max_j |v_j|$, so $|v_\iota| > 1$. By the KKT conditions, $|v_\iota| = |\sum_{j \neq \iota} K_{\iota j}^{-1} v_j / K_{\iota \iota}^{-1}|$. Since $K^{-1} = L + \varepsilon I$ is diagonally dominant, $\sum_{j \neq \iota} |K_{\iota j}^{-1}| / |K_{\iota \iota}^{-1}| \leq 1$. By irreducibility of K^{-1} , all components must have equal magnitude > 1 . But then $v/|v_1|$ is feasible with smaller objective, contradicting optimality. Hence the two optima coincide.

This means *all training points are support vectors* for Laplacian-based kernels. A PD matrix that is not diagonally dominant does not have this property.

□

7. Learning Theory

PAC learning, the No-Free-Lunch theorem, VC dimension, and large-margin generalisation bounds.

7.1 PAC Learnability

Problem 7.1.

Prove $H = \{h_r : r > 0\}$ with $h_r(x) = \mathbf{1}(\|x\|_2 \leq r)$ is PAC learnable with $m_H(\varepsilon, \delta) \leq \lceil \ln(1/\delta)/\varepsilon \rceil$.

Solution 7.1.

Under realizability, $\exists r^*$ with $f^*(x) = \mathbf{1}(\|x\|_2 \leq r^*)$. The algorithm returns h_{r_S} where $r_S = \max\{\|x_i\|_2 : y_i = 1\} \leq r^*$. Define r_ε such that $P(r_\varepsilon \leq \|x\| \leq r^*) = \varepsilon$. Then:

$$\begin{aligned} P(L_D(h_{r_S}) \geq \varepsilon) &= P(r_S \leq r_\varepsilon) = \prod_{i=1}^m P(\|x_i\| \leq r_\varepsilon \text{ or } \|x_i\| > r^*) \\ &= (1 - \varepsilon)^m \leq e^{-\varepsilon m} \leq \delta \end{aligned}$$

for $m \geq \ln(1/\delta)/\varepsilon$. □

Problem 7.2.

Prove $1 - x \leq e^{-x}$ for all $x \in \mathbb{R}$.

Solution 7.2.

Let $g(x) = e^{-x} + x - 1$. Then $g'(x) = -e^{-x} + 1$ and $g''(x) = e^{-x} > 0$, so g is convex. It attains its minimum at $g'(x) = 0 \Leftrightarrow x = 0$, where $g(0) = 0$. Hence $g(x) \geq 0$ for all x . □

7.2 The No-Free-Lunch Theorem

Problem 7.3.

Generalise the No-Free-Lunch lower bound to $\frac{k-1}{2k}$ when $|X| \geq km$.

Solution 7.3.

Follow the proof of the NFL Theorem. Let $C \subseteq X$ with $|C| = km$. The number of unseen examples from C is $p \geq (k-1)m$. The key step changes from $L_{D_i}(h) \geq \frac{1}{2p} \sum_r \mathbf{1}(h(v_r) \neq f_i(v_r))$ to $L_{D_i}(h) = \frac{1}{km} \sum_{x \in C} \mathbf{1}(h(x) \neq f_i(x)) \geq \frac{k-1}{kp} \sum_{r=1}^p \mathbf{1}(h(v_r) \neq f_i(v_r))$. The factor $1/2$ is replaced by $(k-1)/k$. The rest follows identically, yielding $\max_i \mathbb{E}[L_{D_i}(A(S))] \geq \frac{1}{2} \cdot \frac{k-1}{k} = \frac{k-1}{2k}$. □

7.3 VC Dimension

Problem 7.4.

Find the VC-dimension of (a) $H_{=k}^X$ and (b) $H_{at-most-k}$.

Solution 7.4.

(a) $\text{VCdim}(H_{=k}^X) = \min\{k, |X| - k\}$.

Lower bound: A set of $\ell = \min\{k, |X| - k\}$ points can be shattered because for any labelling with $N_+ \leq \ell \leq k$ positive labels, we can assign label 1 to $k - N_+$ additional points from $X \setminus C$.

Upper bound: A set of $k + 1$ all-positive points cannot be realised (only k positives allowed). A set of $|X| - k + 1$ all-negative points cannot be realised either. So $\text{VCdim} < 1 + \min\{k, |X| - k\}$.

(b) $\text{VCdim}(H_{at-most-k}) = \min\{2k + 1, |X|\}$.

Lower bound: A set of $2k + 1$ points can be shattered since any labelling has either $N_+ \leq k$ or $N_- \leq k$.

Upper bound: For $2k + 2$ points, the labelling with $k + 1$ positives and $k + 1$ negatives cannot be realised. $\square$

Problem 7.5.

(a) Infinite H with $\text{VCdim} = 1$. (b) Finite H with $\text{VCdim} = \lfloor \log_2 |H| \rfloor$.

Solution 7.5.

(a) $H = \{h_y : y \in [0, 1]\}$ with $h_y(x) = \mathbf{1}(x \leq y)$. The point $z = 0.5$ is shattered (use h_1 for label 1, h_0 for label 0). But no two points $z_1 < z_2$ can be shattered: the labelling $(0, 1)$ requires $h_y(z_1) = 0 \Rightarrow y < z_1$, but then $h_y(z_2) = 0 \neq 1$.

(b) $H = \{h_{1/4}, h_{3/4}\}$, so $|H| = 2$ and $\lfloor \log_2 2 \rfloor = 1$. The point $z = 0.5$ is shattered. $\square$

7.4 Large-Margin Classifiers

Problem 7.6.

Why formulate VC-dim of large margin classifiers w.r.t. a fixed X ?

Solution 7.6.

If $X = \{x \in \ell_2 : \|x\|_2 \leq R\}$, then there exists $\tilde{x} \in X$ with $\|\tilde{x}\|_2 < \Lambda^{-1}$. By Cauchy-Schwarz, $|\langle w, \tilde{x} \rangle| \leq \|w\|_2 \|\tilde{x}\|_2 < \Lambda \cdot \Lambda^{-1} = 1$, violating the constraint $|\langle w, x \rangle| \geq 1$. So $H_{X, \Lambda}$ would be empty. A fixed finite X avoids this. $\square$

Problem 7.7.

(a) Prove $\text{VCdim}(H_{X, \Lambda}) \leq \Lambda^2 \max_{x \in X} \|x\|_2^2$. (b) Show the bound is tight.

Solution 7.7.

(a) By the Littlestone dimension result ($\text{VC-dim} \leq \text{max mistakes of any deterministic online algorithm}$), apply this to the Perceptron. The Novikoff bound gives at most $(R/\gamma)^2 = \|w\|_2^2 \max_x \|x\|_2^2 \leq \Lambda^2 \max_x \|x\|_2^2$ mistakes.

(b) Fix $\Lambda > 0$ and let $X_\Lambda = \{\tilde{x}\}$ with $\tilde{x} = (\Lambda^{-1}, 0, \dots)$. Then $\Lambda^2 \|\tilde{x}\|_2^2 = 1$. The constraint $|\langle w, \tilde{x} \rangle| \geq 1$ with $\|w\| \leq \Lambda$ forces $\langle w, \tilde{x} \rangle = \pm 1$, so $H_{X_\Lambda, \Lambda} = \{\tilde{x} \mapsto 1, \tilde{x} \mapsto -1\}$, giving $\text{VCdim} = 1 = \lfloor \Lambda^2 \|\tilde{x}\|_2^2 \rfloor$. $\square$